MILITARY SPECIFICATION

PERISCOPE, TANK: M36E1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a tank commander's periscope which functions in both visible and the infrared light regions. The periscope is a part of the fire control system in the 105MM Gun Tank, M60 Series and is used for observation and machine gun sighting.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-B-14837	Body Assembly: 8619920
MIL-E-48441	Elbow Assembly: 11733401
MIL-H-14844	Head Assembly: 8619900
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1240

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DRAWINGS

U.S. Army, Frankford Arsenal

F8565964	Test Fixture for Periscopes M34 and M36
F8575658	Vibration Adapter for Periscopes M34 and M36
F10554769	10" Collimator (8° Field of View)
F11747318	Periscope, Tank: M36E1
F11732167	Gage, Interchangeability, Maximum
F11732168	Gage, Interchangeability, Minimum
F11746731	Test Stand

PACKAGING DATA SHEET

PS11747318	Packaging of Periscope, Tank: M36E1

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The periscope shall be manufactured in accordance with Drawing F11747318 and drawings pertaining thereto. The following shall have met their respective specifications (except that they need not be subjected to separate vibration tests) and be interchangeable within the requirements of this specification.

Body (Daylight) Assembly	8619920	MIL-B-14837
Elbow (Passive) Assembly	11733401	MIL-E-48441
Head Assembly	8619900	MIL-H-14844

3.2.1 Interchangeability.

a. Modules.- The periscope shall meet the requirements of 3.6.2, 3.6.5, 3.6.6 and 3.6.8 without need for realignment of components or selection of modules listed in 3.2. Adjustments, other than those specified herein, are prohibited.

b. Periscope with mount.- The periscope shall be capable of insertion into the gages specified in 4.6.5.2. After insertion, the periscope shall be capable of being firmly secured and subsequently removed.

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3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Vibration.- This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to the contacts for the regulator and the reticle projector lamp (Reference Drawing, F11733401). The periscope shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/6 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the periscope shall show no evidence of physical failure subsequent to vibration. Following the vibration test the periscope shall meet all remaining requirements and then each module (see 3.2) shall meet their own respective post vibration test requirements.

3.5 Cleanliness.- With the periscope disassembled, the three module units shall meet the following requirements.

a. Defects.- There shall be no evidence of moisture, grease, finger prints, condensates, fractures, and adhesive separations on any glass component when viewing through either end of each module (including the unity power system).

b. Dirt.

(1) Body assembly (daylight).- When viewed through the eyepiece, there shall be no more than three dirt particles appearing within a 50 mil diameter central field of view. There shall be no more than three dirt particles appearing outside the specified field of view. The angular separation between dirt particles shall be at least 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution. The size of any particle shall not exceed the apparent width of the reticle line.

(2) Elbow assembly (passive).- When viewing through the eyepiece with the image intensifier tube energized, there shall be no more than five particles visible on the screen surface. The size of any particle shall not exceed the apparent width of the reticle line.

(3) Head assembly.- There shall be no more than three particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05 inch.

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3.6 Performance.

3.6.1 Orientation.- Datum Plane "B" (Reference Drawing F8619900) shall lie in a vertical plane. The elevation input arm shall be positioned as indicated on Drawing F8619900. The Datum Plane "A" (F8619900) shall be in a horizontal plane.

3.6.2 Parallax.- Parallax between the reticle and the image of a target at 1200 meters shall not exceed 0.2 mil.

3.6.3 Collimation.- These requirements shall be met prior to meeting the requirements of 3.4.

a. An incident line of sight which is perpendicular within 0.2 mil to Datum Plane "B".

b. The elevation input arm positioned in accordance with Drawing F8619900.

3.6.3.1 Pre-vibration setting.- Rotate daylight and passive boresight knobs to establish coincidence of each boresight cross with the incoming line of sight (see 3.6.3a) prior to meeting requirements of 3.4.

3.6.3.1.1 Daylight viewing area.

a. Elevation knob travel.- The boresight knob shall provide a minimum travel of 5.0 mils up and 8.0 mils down from the collimation setting of 3.6.3a.

b. Deflection knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.6.3a.

c. Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.6.3a, shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.6.3.1.2 Passive viewing area.

a. Elevation knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in elevation and depression from the collimated setting of 3.6.3a.

b. Deflection knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.6.3a.

c. Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.6.3a shall not exceed 0.3 mil (total variation not to exceed 0.6 mil).

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3.6.3.2 Post vibration collimation.- The original collimation settings established in 3.6.3.1 and 3.6.3.1.1 shall not deviate more than 0.3 mil (including fixture error) as observed through eyepiece subsequent to meeting requirements of 3.4.

3.6.4 Collimation repeatability.

3.6.4.1 Assembling daylight body to head assembly.- The daylight body assembly shall be capable of repeating an original line of sight within 0.20 mil when removed and reinstalled (a minimum of 3 times) into the head assembly.

3.6.4.2 Assembling passive elbow to daylight body/head assembly.- The passive elbow assembly shall be capable of repeating an original line of sight within 0.20 mil when removed and reinstalled (a minimum of 3 times) into the combination daylight body and head assembly.

3.6.5 Image tilt - daylight and passive.- The image of a target line which is originally parallel to the intersection of Datum Planes "A" and "B" shall emerge parallel to this same intersection within 2.0 degrees of arc.

3.6.6 Parallelism of reticle and image - daylight and passive.- A line originally parallel to the intersection of Datum Plane "A" and "B" shall be parallel to the horizontal reticle line within 2.0 degrees of arc.

3.6.7 Objective focusing range - passive.- Rotation of Ring 11733436 shall cause the objective to bring targets from 50 meters to 3000 meters into focus.

3.6.8 Resolution.

3.6.8.1 Daylight.- The resolution shall be 10 seconds of arc or less in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.6.8.2 Passive.- The resolution shall be 1 minute of arc or less in two meridians. Eyepiece focus shift between the meridian shall not exceed 0.25 diopter.

3.6.9 Electrical characteristics - passive.

3.6.9.1 Image intensifier tube.- The image intensifier tube shall be energized by the application of +24 $\pm$ 6.0 VDC to the contact for the regulator (Ref. Dwg. F11733401).

3.6.9.2 Image steadiness.- There shall be no visible flicker, no movement of the image nor audible sparking when observed for 1 minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

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3.6.9.3 Reticle illumination.- With a minimum of 18 and a maximum of 30 volts applied to the contact for the reticle lamp (Ref. Dwg. F11733401) the reticle marking shall be visible when viewing through the eyepiece with the image tube energized (see 3.6.9.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three periscopes plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for meterials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP listed in the technical data package (TDP)). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.2, the tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Line of sight variation, daylight viewing area	3.6.3.1.1c	4.6.4.3.3
2.	Line of sight variation, passive viewing area	3.6.3.1.2c	4.6.4.4.3
3.	Resolution, daylight viewing area	3.6.8.1	4.6.4.8
4.	Resolution, passive viewing area	3.6.8.2	4.6.4.9
5.	Image intensifier tube	3.6.9.1	4.6.4.1.1
6.	Image steadiness	3.6.9.2	4.6.4.1.2
7.	Reticle illumination	3.6.9.3	4.6.4.1.3
8.	Interchangeability, periscope with mount	3.2.1b	4.6.5.2

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between +60° and +90°F.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS
CRITICAL: NONE
MAJOR: AQL 0.65% Defective

NO.	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
101.	Pre-vibration collimation setting	3.6.3.1	4.6.1
102.	Vibration (Periscope)	3.4	4.6.2
103.	Post-vibration collimation	3.6.3.2	4.6.2.1 & 4.6.2.1.1

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TABLE I - CLASSIFICATION OF DEFECTS CON'T

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
104.	Cleanliness	3.5	4.6.3
105.	Parallax	3.6.2	4.6.4.2
106.	Elevation knob travel, daylight viewing area	3.6.3.1.1a	4.6.4.3.1
107.	Deflection knob travel, daylight viewing area	3.6.3.1.1b	4.6.4.3.2
108.	Elevation knob travel, passive viewing area	3.6.3.1.2a	4.6.4.4.1
109.	Deflection knob travel, passive viewing area	3.6.3.1.2b	4.6.4.4.2
110.	Collimation repeatability assembling daylight body to head assembly	3.6.4.1	4.6.2.1.2.1
111.	Collimation repeatability, assembling passive elbow to daylight body/head assembly	3.6.4.2	4.6.2.1.2.2
112.	Image tilt, daylight and passive	3.6.5	4.6.4.5
113.	Parallelism of reticle and image, daylight and passive	3.6.6	4.6.4.6
114.	Objective focusing range, passive	3.6.7	4.6.4.7

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One periscope shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable drawings and specifications - visual
302.	General specification	3.3	MIL-F-13926 - visual

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4.4.1.1 Three periscopes that have met the requirements and tests in 4.3.2.2 and in Table I shall be selected at random as a special sample from each months production. Each sample shall then meet the requirement and test in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Interchangeability, modules	3.2.1a	4.6.5.1

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SOAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Special inspection equipment.

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4.5.2.3.1 Performance.- Special testing equipment to test performance of the complete periscope shall conform to the following requirements:

a. A Collimator conforming to Drawing F10554769 (10" Collimator (8° Field of View)) shall be positioned on a stand conforming to Drawing F11746731 (Test Stand). Collimator and Test Stand shall be set-up in accordance with the set-up instruction on their respective drawings. The Collimator shall be calibrated in accordance with the calibration instructions on the drawing. Light assembly "C" shall be utilized for performing test for daylight characteristics. When performing test for passive characteristics light assembly "Y" shall be utilized.

b. A holding medium conforming to Drawing F8565964 (Test Fixture for Periscope M34 and M36) shall be positioned on a table having a flat and level top such that the line of sight of the periscope is coincident with that of the collimator.

c. A calibrated observation telescope of at least three power (3x) shall be a part of the test equipment. The observation telescope shall be capable of being positioned and mounted so that the optical axis of the telescope is in alignment with the line of sight emitting from the eyepiece of the periscope. The reticle of the telescope shall have a crossline reticle pattern with tolerance lines (3.6.5) to determine image tilt. The vertical reticle line of the telescope shall be adjusted plumb.

d. A power supply for energizing the image tube and the reticle projector of the periscope (see 3.6.9.1) shall be a part of the testing equipment. The power supply shall be capable of supplying and maintaining the required voltages.

e. The observation telescope shall be capable of being removed from the optical line of sight of the periscope when not required in the performance tests.

4.6 Test methods and procedures.

NOTE: Caution shall be exercised during the following tests to prevent Arm Assembly 8635116 and Arm 8619845 from striking hard against their respective stop pins.

4.6.1 Pre-vibration collimation setting.- Prior to the vibration test, the periscope shall be positioned as in 4.6.4 to determine compliance with 3.6.3 and 3.6.3.1. While viewing through each periscope eyepiece (daylight and passive), obtain coincidence of the periscope reticle boresight crosses and the projected target image crossline, by means of the daylight and passive boresight knobs (do not slip knob scales). When coincidence has been obtained, record the boresight knob scale settings for use in the test in 4.6.2.1.1. Without disturbing the knob scale settings, the periscope shall be then subjected to the vibration test in 4.6.2.

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NOTE: Insure that knob scales are not disturbed (re-slipped) from their original recorded settings throughout remainder of tests herein.

4.6.2 Vibration (periscope).- Inspection equipment capable of providing the amplitude and frequency specified in 3.4 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The periscope shall be clamped on the vibrating equipment by means of an adapter that conforms to Drawing F8575658, Vibration Adapter for Periscopes M34 and M36. The vibration motion shall be in the vertical plane. During this test, the elevation input arm shall be immobilized to any position that shall prevent Arm Assembly 8635116 and Arm 8619485 from contacting their respective stop pins. Prior to vibration, energize the image intensifier tube and reticle projector lamp as specified in 3.4. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece (passive). The periscope shall then be vibrated for the time duration specified in 3.4. During the first, third and fifth minutes of the vibration cycle, observe the screen surface of the tube and the reticle pattern. During these observations, there shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp.

NOTE: When provided as part of a complete periscope, vibration of each head, daylight body and elbow (passive) assembly, including their respective post-vibration collimation test, will not be required. A record of all data (collimation, pre-vibration setting, etc.) necessary to determine compliance with required performance shall be available.

4.6.2.1 Post-vibration.- Immediately following vibration, the periscope shall again be positioned and adjusted as in 4.6.4 and then subjected to the post-vibration collimation test of 4.6.2.1.1. Insure that the boresight knob scales are not disturbed from the original setting obtained in 4.6.1.

4.6.2.1.1 Post-vibration collimation.- Observe through each eyepiece assembly (daylight and passive) and adjust by means of the boresight knobs to obtain coincidence of each periscope reticle boresight cross with the target image lines. Boresight knob adjustment shall not exceed the tolerance specified in 3.6.3.2, as measured on the boresight knob scales. Record each collimation setting for use in the tests in 4.6.2.1.2.

4.6.2.1.2 Collimation repeatability.

4.6.2.1.2.1 Assembling daylight body to head assembly.- Perform this test subsequent to test in 4.6.2.1.1. Remove the daylight body from the head assembly, removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each

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installation for repeatability of the daylight collimation setting, recorded in 4.6.2.1.1. The periscope shall meet the requirements specified in 3.6.4.1, as measured on the boresight knobs.

4.6.2.1.2.2 Assembling passive elbow to daylight body/head assembly.- Perform this test subsequent to the test in 4.6.2.1.2.1. Remove the passive elbow from the head assembly. Repeated removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each installation for repeatability of the passive collimation setting, recorded in 4.6.2.1.1. The periscope shall meet the requirements specified in 3.6.4.2, as measured on the boresight knobs.

4.6.2.1.3 Post-vibration examination.- The periscope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4, prior to being subjected to the remaining tests specified herein. A visual examination shall also be made for missing or damaged parts, glass chips foreign matter and bonding (separations, chipping or rupture). An audible examination shall also be made of the periscope for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a periscope defective.

4.6.3 Cleanliness.- Subsequent to the post-vibration tests, each separate assembly (see 3.2) shall be subjected to the tests in 4.6.3.1, 4.6.3.2 and 4.6.3.3 respectively to determine compliance with 3.5.

4.6.3.1 Body assembly (daylight).- The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. The body assembly (daylight) shall conform to the requirements of 3.5a and 3.5b(1).

4.6.3.2 Elbow assembly (passive).- The image intensifier tube shall be de-energized for this portion of the test. With 2 each 15 watt desk fluorescent lamp held close to the objective window and eyelens, view by reflected light the glass components of the elbow assembly to determine compliance with the cleanliness requirements of 3.5a. Upon successful completion of this protion of the test, energize the image intensifier tube from a power source capable of supply and maintaining 24 volts DC. The image tube shall be permitted a stabilization period lf 15 seconds prior to performing the cleanliness test. Subsequent to the tube stabilization period, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the eyepiece diopter ring. The screen surface of the image tube shall then be visually inspected for cleanliness in accordance with 3.5b(2) and shall conform to the requirements specified thererin. "Cold" and "Hot" spots on the screen surface of the image tube shall not be considered dirt.

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4.6.3.3 Head assembly.- The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the exit window, inspection for dirt shall be made by viewing the illuminated field specified in 4.6.3.1. The head assembly shall conform to the requirements of 3.5a and 3.5b(3).

4.6.4 Performance.- The following tests shall be performed with the complete periscope mounted in conformance with 3.6.1 and in accordance with special testing equipment conforming to 4.5.2.3.1. Unless otherwise specified, the image tube and reticle projector shall be energized; the Passive Focusing Ring 11733436 shall be adjusted for best focus of the target image after the diopter ring of the passive eyepiece is set for best focus of the phospher grain on the screen of the image tube; the diopter adjustment knob of the daylight eyepiece shall be set for best focus of the periscope reticel; the sector lever of Fixture 8565964 shall be engaged in the zero degree elevation position on the index plate and the Collimator 10554769 shall be adjusted for 1200 meters.

4.6.4.1 Electrical characteristics - passive.

4.6.4.1.1 Image intensifier tube.- Apply the voltage specified in 3.6.9.1 to the contact for the regulator (Reference Drawing F11733401) and position the electrical switch in the "ON" position. After allowing a 15 second tube stabilization period, view through the passive eyepiece and observe the image tube screen. The screen surface of the tube shall be illuminated.

4.6.4.1.2 Image steadiness.- Perform this test with the reticle projector lamp off. Position an observation telescope at the passive eyepiece of the periscope. The telescope shall contain a crossline reticle pattern and be capable of focusing on the screen of the image tube. With the aid of the telescope, view through the passive eyepiece and establish the telescope reticle crosslines coincident to the projected target image crosslines. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker, no movement of the image nor audible sparking as specified in 3.6.9.2. Image movement shall be determined by means of the target image with respect to the observation telescope reticle pattern. Sparking shall be determined by means of an audio examination. Upon completion of this test, energize the reticle projector lamp.

4.6.4.1.3 Reticle illumination.- Perform this test concurrent with the test in 4.6.4.1.2 and while the reticle projector lamp is energized at each voltage specified in 3.6.9.3. The reticle markings shall appear clearly defined at each required voltage setting when visually inspected through the passive eyepiece.

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4.6.4.2 Parallax.- View through the daylight eyepiece and establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While viewing through the eyepiece, parallax will be recognized by any apparent displacement of the intersecting point of the periscope reticle boresight cross with respect to the intersecting point of the projected target image crosslines when the observer's head is moved from side to side or up and down. Displacement between the periscope reticle and target image shall not exceed the tolerance specified in 3.6.2.

4.6.4.3 Daylight viewing area.

4.6.4.3.1 Elevation knob travel.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as specified in 3.6.3.1.1a as measured on the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall travel as specified in 3.6.3.1.1a as measured on the target.

4.6.4.3.2 Deflection knob travel.- Perform this test utilizing the procedure in 4.6.4.3.1 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.6.3.1.1b as measured on the target.

4.6.4.3.3 Line of sight variation.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.6.3.1.1c. Re-establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotate the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the values specified in 3.6.3.1.1c to determine overall conformance with the requirements specified therein.

4.6.4.4 Passive viewing area.

4.6.4.4.1 Elevation knob travel.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as specified in 3.6.3.1.2a

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as measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall travel as specified in 3.6.3.1.2a as measured on the target.

4.6.4.4.2 Deflection knob travel.- Perform this test utilizing the procedure in 4.6.4.4.1 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.6.3.1.2b as measured on the target.

4.6.4.4.3 Line of sight variation.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.6.3.1.2c. Re-establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotate the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the values specified in 3.6.3.1.2c to determine overall conformance with requirements specified therein.

4.6.4.5 Image tilt - daylight and passive.- The daylight and passive optical systems of the periscope shall be inspected utilizing an auxiliary telescope containing a crossline reticle pattern. The vertical reticle line of the auxiliary telescope shall be aligned plumb when in the test position on a qualified level surface. With aid of the auxiliary telescope, view through the passive eyepiece of the periscope and establish one extremity of the telescope vertical reticle line coincident to the vertical target image line. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.6.5, as measured on the target. Repeat the above procedure with the auxiliary telescope positioned at the daylight eyepiece of the periscope. The images formed by the two optical systems (daylight and passive) of the periscope shall meet the requirements specified in 3.6.5.

4.6.4.6 Parallelism of reticle and image - daylight and passive.- Perform this test while viewing through the passive eyepiece of the periscope with the unaided eye. Position one extremity of the periscope horizontal reticle line coincident to the horizontal target image line. The periscope horizontal reticle line shall be parallel to the horizontal target line within the tolerance specified in 3.6.6, as measured on the target. The above procedure shall also be repeated at the daylight eyepiece of the periscope in compliance with 3.6.6.

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4.6.4.7 Objective focusing range - passive.- This portion of the test shall be performed with the 10" collimator adjusted for 50 meters. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, rotate focusing ring of the elbow assembly for best focus of target image to determine compliance with 3.6.7. Upon satisfactory completion of this portion of the test, the 10" collimator shall be readjusted for infinity to determine compliance with the 3,000 meters specified in 3.6.7. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, again rotate the focusing ring of the elbow assembly for best focus of the target image to determine overall compliance with 3.6.7.

4.6.4.8 Resolution - daylight.- The day light optical system of the periscope shall be inspected for resolution utilizing a calibrated dioptometer with a magnification of at least 3 power (3x) and a resolving power chart or a wall target representing the angular subtense for minutes of arc specified in 3.6.8.1. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and contain four lines sets as shown on Figure I therein. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter adjustment knob of the periscope daylight eyepiece to indicate zero diopter. Position the dioptometer at the periscope daylight eyepiece. While viewing through the dioptometer, adjust the objective scale of the dioptometer for best focus of the resolution target. Any change in diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.6.8.1.

4.6.4.9 Resolution - passive.- This test shall be performed while viewing through the passive eyepiece with the unaided eye. Best focus of both meridians of resolution pattern appearing in the first quadrant shall be obtained by adjusting the passive eyepiece diopter ring. Any change in the diopter readings by adjustment of the passive eyepiece diopter ring to obtain the detectable line structure and proper line count in each meridian shall be recorded, as measured on the ring scale. The difference between the readings for the meridians shall not exceed the value specified in 3.6.8.2.

4.6.5 Interchangeability.

4.6.5.1 Modules.- Each sample periscope (see 4.4.1.1) shall be disassembled (removal of passive elbow from daylight body and the daylight body from head) to permit interchange of the component modules. The modules shall be identified and interchanged as shown in Table IV.

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TABLE IV

Original Module Positions

	Periscope 1	Periscope 2	Periscope 3
Head Assembly (HA)	HA-1	HA-2	HA-3
Body Assembly (DL) (Daylight)	DL-1	DL-2	DL-3
Elbow Assembly (P) (Passive)	P-1	P-2	P-3

Interchanged Module Positions

	Periscope 1	Periscope 2	Periscope 3
Head Assembly (HA)	HA-1	HA-2	HA-3
Body Assembly (DL) (Daylight)	DL-2	DL-3	DL-1
Elbow Assembly (P) (Passive)	P-3	P-1	P-2

Upon reassembly of the three interchanged periscopes, each periscope shall be subjected to the tests specified in 4.6.4.2, 4.6.4.5, 4.6.4.6, 4.6.4.8 and 4.6.4.9 to determine compliance with 3.2.1a.

4.6.5.2 Periscope with mount.- This test shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings F11732167 and F11732168 respectively, to determine compliance with 3.2.1b.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11747318. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Periscope, M36E1 is a tank commander's periscope primarily intended for use in the fire control system of the 105MM Gun, Tank M60 Series, and is used for machine gun sighting and observing. The periscope has an 8 power passive system for low light level use and a 7 power daylight system for high light level use.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable stock number.
d. Provisions for first article testing.

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6.3 Elevation shaft.- At no time shall the elevation input arm assemblies be allowed to strike hard against the travel stops.

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